Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The maximum area of a right angled triangle with hypotenuse h is:

(A) $\frac{h^2}{2\sqrt{2}}$

(B) $\frac{h^2}{2}$

(C) $\frac{h^2}{\sqrt{2}}$

(D) $\frac{h^2}{4}$

Q.2

The cost of running a bus from A to B, is Rs. $\left(av + \frac{b}{v}\right)$, where v $\mathrm{km/h}$ is the average speed of the bus. When the bus travels at 30 $\mathrm{km/h}$, the cost comes out to be Rs. 75 while at 40 $\mathrm{km/h}$, it is Rs. 65. Then the most economical speed (in $\mathrm{km/h}$) of the bus is:

(A) 45

(B) 50

(C) 60

(D) 40

Q.3

A point on the hypotenuse of a triangle is at distance a and b from the sides of the triangle. Then the minimum length of the hypotenuse is

(A) $\left(a^{\frac{3}{2}} + b^{\frac{3}{2}}\right)^{\frac{2}{3}}$

(B) $\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{3}{2}}$

(C) $\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{3}$

(D) $\left(a^{\frac{3}{2}} + b^{\frac{3}{2}}\right)^{3}$

Q.4

The curve $\mathrm{y} = \mathrm{xe}^{\mathrm{x}}$ has minimum value equal to

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(A) $-\frac{1}{\mathrm{e}}$

(B) $\frac{1}{\mathrm{e}}$

(C) - e

(D) e

Q.5

What is the minimum value of $\mathrm{px}+\mathrm{qy}\ (p>0, q>0)$ when $xy=r^2$?

(A) $2\mathrm{r}\sqrt{\mathrm{pq}}$

(B) $2\mathrm{pq}\sqrt{\mathrm{r}}$

(C) $-2\mathrm{r}\sqrt{\mathrm{pq}}$

(D) 2 rpq

Q.6

Maximum value of the function $f\left(x\right)=x^{\frac{1}{x}}$ is _____.

(A) $\frac{1}{e}$

(B) $\frac{2}{e}$

(C) e

(D) $e^{\frac{1}{e}}$

Q.7

The largest area of a trapezium inscribed in a semicircle of radius R, if the lower base is on the diameter, is

(A) $\frac{3\sqrt{3}}{4}R^2$

(B) $\frac{\sqrt{3}}{2}R^2$

(C) $\frac{3\sqrt{3}}{8}R^2$

(D) R^2

Q.8

A stone thrown vertically upward satisfies the equation $\mathrm{s}=64\mathrm{t}-16\mathrm{t}^2$, where s is in meter and t is in second. What is the time required to reach the maximum height?

(A) 1 s

(B) 2 s

(C) 3 s

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(D) $4\ \text{s}$

Q.9

Find the minimum value of the function $\frac{40}{3x^4+8x^3-18x^2+60}$

(A) $\frac{1}{3}$

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{5}{3}$

Q.10

A wire $34\ \text{cm}$ long is to be bent in the form of quadrilateral of which each angle is 90°. What is the maximum area which can be enclosed inside the quadrilateral?

(A) $68\ \text{cm}^2$

(B) $70\ \text{cm}^2$

(C) $71.25\ \text{cm}^2$

(D) $72.25\ \text{cm}^2$

Q.11

If the function $y = \frac{ax+b}{(x-1)(x-4)}$ has turning point at $P(2, -1)$, then

(A) $a = b = 1$

(B) $a = b = 0$

(C) $a = 1, b = 0$

(D) $a = b = 2$

Q.12

If $f(x) = x \ln x$, then $f(x)$ attains minimum value at which one of the following points?

(A) $x = e^{-2}$

(B) $x = e$

(C) $x = e^{-1}$

(D) $x = 2e^{-1}$

Q.13

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The profit function, in rupees, of a firm selling x items $(x \geq 0)$ per week is given by $\mathrm{P}(\mathrm{x})=-3500+(400-\mathrm{x})\mathrm{x}$. How many items should the firm sell so that the firm has maximum profit?

(A) 400

(B) 300

(C) 200

(D) 100

Q.14

The fuel charges for running a train are proportional to the square of the speed generated in miles per hour and costs Rs. 48 per hour at 16 miles per hour. The most economical speed if the fixed charges i.e. salaries etc. amount to Rs. 300 per hour is

(A) 10

(B) 20

(C) 30

(D) 40

Q.15

The difference between greatest and least value of $f(x)=2 \sin x+\sin 2x, x \in\left[0, \frac{3 \pi}{2}\right]$ is

(A) $\frac{3 \sqrt{3}}{2}$

(B) $\frac{3 \sqrt{3}}{2}-2$

(C) $\frac{3 \sqrt{3}}{2}+2$

(D) None of these

Q.16

A rod AB of length $16 \mathrm{~cm}$. rests between the wall AD and a smooth peg, $1 \mathrm{~cm}$ from the wall and makes an angle θ with the horizontal. The value of θ for which the height of G, the mid point of the rod above the peg is minimum, is

(A) 15°

(B) 30°

(C) 60°

(D) 75°

Q.17

What is the product of two parts of 20 , such that the product of one part and the cube of the other is maximum?

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(A) 75

(B) 91

(C) 84

(D) 96

Q.18

If $\mathrm{A} > 0,\ \mathrm{B} > 0$ and $\mathrm{A} + \mathrm{B} = \pi/3$, then the maximum value of $\tan A \tan B$ is

(A) $\frac{1}{\sqrt{3}}$

(B) $\frac{1}{3}$

(C) 3

(D) $\sqrt{3}$

Q.19

Let $f : [a, b] \to R$ be a function such that for $c \in (a, b)$, $f'(c) = f'(c) = f''(c) = f^{iv}(c) = f^{v}(c) = 0$. Then

(A) f has a local extremum at $x = c$

(B) f has neither local maximum nor minimum a $x = c$

(C) f is necessarily a constant function

(D) it is difficult to say whether (a) or (b)

Q.20

Let $f(x)$ be a function defined as follows: $f(x) = \sin\left(x^2 - 3x\right), x \leq 0$; and $6x + 5x^2, x > 0$ Then at $x = 0, f(x)$

(A) has a local maximum

(B) has a local minimum

(C) is discontinuous

(D) None of these

Answers & Solutions

Q.1 Answer:

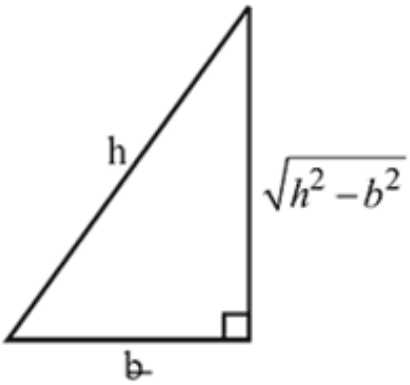
$\frac{h^2}{4}$

Solution:

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Let base $=b$

Altitude (or perpendicular) $=\sqrt{h^2-b^2}$



Area, $\mathrm{A}=\frac{1}{2}\times$ **base** $\times$ **altitude**

$=\frac{1}{2}\times b\times\sqrt{h^2-b^2}$

$\Rightarrow \frac{d\,\mathrm{A}}{db}=\frac{1}{2}\left[\sqrt{h^2-b^2}+b\cdot\left(\frac{-2b}{2\sqrt{h^2-b^2}}\right)\right]$

$=\frac{1}{2}\left[\frac{h^2-2b^2}{\sqrt{h^2-b^2}}\right]$

Put $\frac{d\,\mathrm{A}}{db}=0, \Rightarrow \quad b=\frac{h}{\sqrt{2}}$

Maximum area $=\frac{1}{2}\times\frac{h}{\sqrt{2}}\times\sqrt{h^2-\frac{h^2}{2}}=\frac{h^2}{4}$

Q.2 Answer:

60

Solution:

Let cost $\mathrm{C}=av+\frac{b}{v}$

According to given question, $30a+\frac{b}{30}=75\cdots$ **(i)**

$40a+\frac{b}{40}=65$ **...(ii)**

On solving (i) and (ii), we get

$a=\frac{1}{2}$ **and** $b=1800$ **Now,** $C=av+\frac{b}{v}\Rightarrow\frac{dC}{dv}=a-\frac{b}{v^2}$

$\frac{dC}{dv}=0\Rightarrow a-\frac{b}{v^2}=0\Rightarrow v=\sqrt{\frac{b}{a}}=\sqrt{3600}$

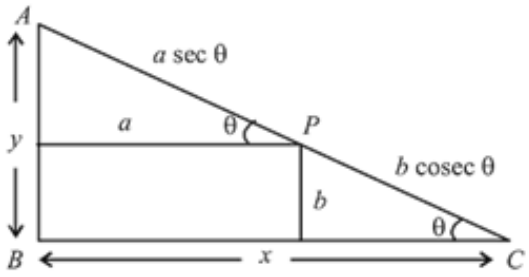
$\Rightarrow v=60\mathrm{kmph}$

Q.3 Answer:

$\left(a^{\frac{2}{3}}+b^{\frac{2}{3}}\right)^{\frac{3}{2}}$

Solution:

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From the figure, $PC = b\,\text{cosec}\,\theta$ **and** $AP = a\sec\theta$

$AC = PC + AP$

or $AC = b\,\text{cosec}\,\theta + a\sec\theta$ **...(1)**

$\therefore \frac{d(AC)}{d\theta} = -b\,\text{cosec}\,\theta\cot\theta + a\sec\theta\tan\theta$

For minimum length, $\frac{d(AC)}{d\theta} = 0$ **or**

$a\sec\theta\tan\theta = b\,\text{cosec}\,\theta\cot\theta$ **or** $\tan\theta = \left(\frac{b}{a}\right)^{1/3}$

$\therefore \sin\theta = \frac{(b)^{1/3}}{\sqrt{a^{2/3}+b^{2/3}}}$ **and**

$\cos\theta = \frac{(a)^{1/3}}{\sqrt{a^{2/3}+b^{2/3}}}$ **...(2)**

Also, $\theta \in (0, \pi/2)$

$\lim_{\theta\to 0}(a\sec\theta + b\,\text{cosec}\,\theta) \to \infty$

and $\lim_{\theta\to\pi/2}(a\sec\theta + b\,\text{cosec}\,\theta) \to \infty$

Therefore, $\theta = \tan^{-1}\left(\frac{b}{a}\right)^{1/3}$ **is a point of minima. For this value of** θ, $AC = \frac{b\sqrt{a^{2/3}+b^{2/3}}}{b^{1/3}} + \frac{a\sqrt{a^{2/3}+b^{2/3}}}{a^{1/3}}$

$[\text{Using}(1)\text{ and }(2)]$

$= \sqrt{a^{2/3}+b^{2/3}}\left(b^{2/3}+a^{2/3}\right) = \left(a^{2/3}+b^{2/3}\right)^{3/2}$

Hence, the minimum length of the hypotenuse is $\left(a^{2/3}+b^{2/3}\right)^{3/2}$

Q.4 Answer:

$-\frac{1}{e}$

Solution:

Let $y = xe^x$

Differentiate both side w.r.t. ' x **'.** $\Rightarrow \frac{dy}{dx} = e^x + xe^x = e^x(1+x)$

Put $\frac{dy}{dx} = 0$

$\Rightarrow \quad e^x(1+x) = 0 \Rightarrow x = -1$

Now, $\frac{d^2y}{dx^2} = e^x + e^x(1+x) = e^x(x+2)$

$\left(\frac{d^2y}{dx^2}\right)_{(x=-1)} = \frac{1}{e} + 0 > 0$

Hence, $y = xe^x$ **is minimum function and** $y_{min} = -\frac{1}{e}$

Q.5 Answer:

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$2r\sqrt{pq}$

Solution:

Given that $xy = r^2$

$\Rightarrow \quad y = \frac{r^2}{x}$

Let $S = px + qy = px + \frac{qr^2}{x}$

$\Rightarrow \frac{dS}{dx} = p - \frac{qr^2}{x^2}$

$\frac{dS}{dx} = 0$ **for maximum or minimum.**

So, $0 = p - \frac{qr^2}{x^2}$

$\Rightarrow \quad x^2 = \frac{qr^2}{p} \Rightarrow x = \pm\sqrt{\frac{q}{p}}r$

Now, $\frac{d^2S}{dx^2} = \frac{2qr^2}{x^3}$

At $x = +\sqrt{\frac{q}{p}}r, \frac{d^2S}{dx^2} > 0$

Hence, S is minimum at $x = \sqrt{\frac{q}{p}} \cdot r$

$\Rightarrow \quad y = \frac{r^2}{\sqrt{\frac{q}{p}} \cdot r} = \sqrt{\frac{p}{q}} \cdot r$

Minimum value of $px + qy = p\sqrt{\frac{q}{p}} \cdot r + q\sqrt{\frac{p}{q}}, r$

$\sqrt{pq}r + \sqrt{pq}r = 2r\sqrt{pq}$

Q.6 Answer:

$e^{\frac{1}{e}}$

Solution:

Let $y = x^{\frac{1}{x}}$

$\ln y = \frac{1}{x}\ln x$

$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x^2} - \frac{1}{x^2}\ln x$

$\frac{dy}{dx} = \frac{(1 - \ln x)}{x^2} . x^{\frac{1}{x}}$

$f'(x) = 0$, **at** $x = e$

$f'x|_{x \to e^+} < 0$ **and** $f'x|_{x \to e^-} > 0$

Thus at $x = e$**, there is a maximum value.**

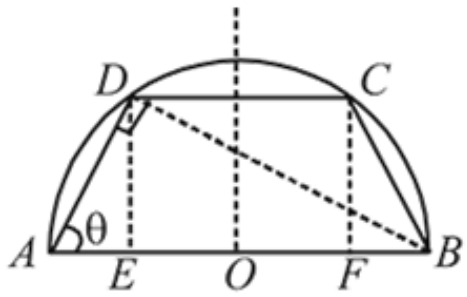
So, maximum value $= e^{\frac{1}{e}}$

Q.7 Answer:

$\frac{3\sqrt{3}}{4}R^2$

Solution:

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$AD = AB\cos\theta = 2R\cos\theta, AE = AD\cos\theta = 2R\cos^2\theta$
or $EF = AB - 2AE = 2R - 4R\cos^2\theta$
$DE = AD\sin\theta = 2R\sin\theta\cos\theta$
Thus, area of trapezium,
$S = \frac{1}{2}(AB + CD) \times DE = \frac{1}{2}\left(2R + 2R - 4R\cos^2\theta\right) \times 2R\sin\theta\cos\theta$
$= 4R^2\sin^3\theta\cos\theta$
$\frac{dS}{d\theta} = 12R^2\sin^2\theta\cos^2\theta - 4R^2\sin^4\theta$
$= 4R^2\sin^2\theta\left(3\cos^2\theta - \sin^2\theta\right)$
For maximum area, $\frac{dS}{d\theta} = 0$ **or** $\tan^2\theta = 3$ **or** $\tan\theta = \sqrt{3}$
$(\theta$ **is acute** $)$ **or** $S_{\max} = \frac{3\sqrt{3}}{4}R^2$

Q.8 Answer:

2 s

Solution:

Given equation is $s = 64t - 16t^2$
$\therefore$ **On differentiating w.r.t.** t**, we get**
$\frac{ds}{dt} = 64 - 32t$
Put $\frac{ds}{dt} = 0$ **for maximum height**
$64 - 32t = 0$
$t = 2$
Now, $\frac{d^2s}{dt^2} = -32$
At $t = 2, \frac{d^2s}{dt^2} = -32$
Since, $\left(\frac{d^2s}{dt^2}\right)_{t=2} < 0$
$\therefore$ **Required time** $= 2$ **second**

Q.9 Answer:

$\frac{2}{3}$

Solution:

Let $y = \frac{1}{40}\left(3x^4 + 8x^3 - 18x^2 + 60\right)$
$\Rightarrow \frac{dy}{dx} = \frac{1}{40}\left(12x^3 + 24x^2 - 36x\right)$

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and $\frac{d^2y}{dx^2} = \frac{1}{40}\left(36x^2 + 48x - 36\right)$

Now $\frac{dy}{dx} = 0 \Rightarrow x^3 + 2x^2 - 3x = 0$

or $x(x-1)(x+3) = 0$ or $\quad x = 0, 1, -3$

At $x = 0$, $\frac{d^2y}{dx^2} = -36 < 0 \therefore y$ is maximum at $x = 0$

$\Rightarrow$ The given function i.e. $\frac{1}{y}$ is minimum at $x = 0$

$\therefore$ minimum value of the function $= \frac{40}{60} = \frac{2}{3}$

Q.10 Answer:

72.25 cm^2

Solution:

Let one side of quadrilateral be x and another side be y

so, $2(x + y) = 34$

or, $(x + y) = 17$ ...(i)

We know from the basic principle that for a given perimeter square has the maximum area, so, $x = y$ and putting this value in equation (i)

$x = y = \frac{17}{2}$

Area $= x \cdot y = \frac{17}{2} \times \frac{17}{2} = \frac{289}{4} = 72.25$

Q.11 Answer:

$a = 1, b = 0$

Solution:

$y = \frac{ax+b}{(x-1)(x-4)} = \frac{ax+b}{x^2-5x+4}$ has turning point at $P(2, -1)$. Thus, $P(2, -1)$ lies on the curve.

Therefore,

$2a + b = 2$ ...(1)

Also, $\frac{dy}{dx} = 0$ at $P(2, -1)$.

Now, $\frac{dy}{dx} = \frac{a\left(x^2-5x+4\right)-(2x-5)(ax+b)}{(x^2-5x+4)^2}$

at $P(2, -1)$, $\frac{dy}{dx} = \frac{-2a+2a+b}{4} = 0$

or $b = 0$ or $a = 1$ [From equation (1)]

Q.12 Answer:

$x = e^{-1}$

Solution:

Let $f(x) = x \ln x$

$f'(x) = \frac{x}{x} + \ln x = 1 + \ln x$

Put $f'(x) = 0 \Rightarrow 1 + \ln x = 0$

$\Rightarrow \ln x = -1 \Rightarrow x = e^{-1}$

Now, $f''(x) = \frac{1}{x}$

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$f''(x)|_{x=e^{-1}} = \frac{1}{e^{-1}} = e > 0$

Hence, $f(x)$ attains minimum value at $x = e^{-1}$

Q.13 Answer:

200

Solution:

$P(x) = -3500 + (400 - x)x = -3500 + 400x - x^2$

On differentiating w.r.t. x, we get

$P'(x) = 400 - 2x$

Put $P'(x) = 0$ for maxima or minima

$\Rightarrow \quad 400 - 2x = 0$

$\Rightarrow \quad x = 200$

Now $P''(x) = -2$

$\Rightarrow \quad P''(200) = -2 < 0$

$\therefore P(x)$ **is maximum at $x = 200$**

Hence 200 items should the firm sell so that the firm has maximum profit.

Q.14 Answer:

40

Solution:

Let the speed of the train be v and distance to be covered be s so that total time taken is $\mathrm{s/v}$ hours.

Cost of fuel per hour $= \mathrm{kv}^2 (\mathrm{k}$ is constant $)$ Also $48 = \mathrm{k}.16^2$ by given condition

$\mathrm{k} = \frac{3}{16}$

Cost to fuel per hour $\frac{3}{16}\mathrm{v}^2$.

Other charges per hour are 300 .

Total running cost,

$C = \left(\frac{3}{16}v^2 + 300\right)\frac{s}{v} = \frac{3s}{16}v + \frac{300s}{v}$

$\frac{\mathrm{dC}}{\mathrm{dy}} = \frac{3\,\mathrm{s}}{16} - \frac{300\,\mathrm{s}}{\mathrm{v}^2} = 0 \Rightarrow \mathrm{v} = 40$

$\frac{\mathrm{d}^2\mathrm{C}}{\mathrm{dv}^2} = \frac{600\,\mathrm{s}}{\mathrm{v}^3} > 0 \therefore \mathrm{v} = 40$ **results in minimum running cost.**

Q.15 Answer:

$\frac{3\sqrt{3}}{2} + 2$

Solution:

$f(x) = 2\sin x + \sin 2x$

$f'(x) = 2\cos x + 2\cos 2x = 2(\cos x + \cos 2x)$

$f'(x) = 0 \Rightarrow 2\cos^2 x + \cos x - 1 = 0$

$\cos x = \frac{-1 \pm 3}{4} = -1, \frac{1}{2} \quad \therefore x = \pi, \frac{\pi}{3}$

Now, $f(0) = 0, f\left(\frac{3\pi}{2}\right) = -2$

$\mathrm{f}(\pi) = 0, \mathrm{f}\left(\frac{\pi}{3}\right) = 2\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$

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$\therefore$ Difference between greatest value and least value

$= \frac{3\sqrt{3}}{2} + 2$

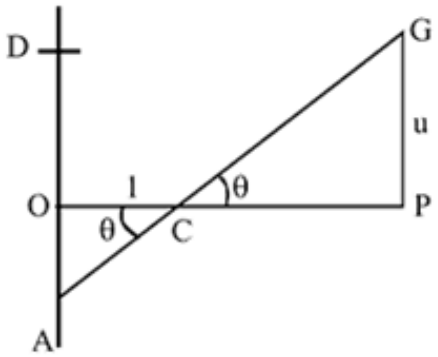
Q.16 Answer:

60°

Solution:

We have $AC = \sec\theta, AG = 8$

$\therefore CG = 8 - \sec\theta$ (C **being the peg**)



But $u = CG\sin\theta = (8 - \sec\theta)\sin\theta$

$u = 8\sin\theta - \tan\theta$

$\frac{du}{d\theta} = 8\cos\theta - \sec^2\theta$

$\frac{d^2u}{d\theta^2} = -8\sin\theta - 2\sec^2\theta\tan\theta$

$\frac{du}{d\theta} = 0$, **when** $\cos^3\theta = \frac{1}{8}, \cos\theta = \frac{1}{2}$

$\frac{d^2u}{d\theta^2} > 0\,(\text{at}\,\theta = 60^\circ), \quad \therefore \theta = 60^\circ$

Q.17 Answer:

75

Solution:

Let 20 be divided in two parts such that

first part $= x$

$\therefore$ **Second part** $= 20 - x$

Now, assume that

$P = x^3(20 - x) = 20x^3 - x^4$

Now, $\frac{dP}{dx} = 60x^2 - 4x^3$; **and** $\frac{d^2P}{dx^2} = 120x - 12x^2$

Put $\frac{dP}{dx} = 0$ **for maxima or minima**

$\Rightarrow \quad \frac{dP}{dx} = 0$

$\Rightarrow \quad 4x^2(15 - x) = 0 \Rightarrow \quad x = 0, x = 15$

$\therefore \left(\frac{d^2P}{dx^2}\right)_{x=15} = 120 \times 15 - 12 \times (225)$

$= 1800 - 2700 = -900 < 0$

$\therefore \quad P$ **is a maximum at** $x = 15.$

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$\therefore$ **First part** $= 15$ **and second part** $= 20 - 15 = 5$

Required product $= 15 \times 5 = 75$

Q.18 Answer:

$\frac{1}{3}$

Solution:

We have, $\mathrm{A} + \mathrm{B} = \frac{\pi}{3}$

$B = \frac{\pi}{3} - A \Rightarrow \tan B = \frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A}$

Let $Z = \tan$ **A. tan B. Then,**

$Z = \tan A \cdot \frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A} = \frac{\sqrt{3}\tan A - \tan^2 A}{1 + \sqrt{3}\tan A}$

$\Rightarrow Z = \frac{\sqrt{3}x - x^2}{1 + \sqrt{3}x}$, **where** $x = \tan A$

$\Rightarrow \frac{dZ}{dx} = -\frac{(x + \sqrt{3})(\sqrt{3}x - 1)}{(1 + \sqrt{3}x)^2}$

For $\max Z$, $\frac{dZ}{dx} = 0 \Rightarrow x = \frac{1}{\sqrt{3}}, -\sqrt{3}$

$\mathrm{x} \neq -\sqrt{3}$ **because** $\mathrm{A} + \mathrm{B} = \pi/3$ **which implies that** $\mathrm{x} = \tan \mathrm{A} > 0.$

It can be easily checked that $\frac{\mathrm{d}^2 Z}{\mathrm{dx}^2} < 0$ **for** $\mathrm{x} = \frac{1}{\sqrt{3}}$**. Hence,** Z **is maximum**

for $x = \frac{1}{\sqrt{3}}$ **i.e.** $\tan \mathrm{A} = \frac{1}{\sqrt{3}}$ **or** $\mathrm{A} = \pi/6.$

For this value of x, $Z = \frac{1}{3}$.

Q.19 Answer:

it is difficult to say whether (a) or (b)

Solution:

For $f(x) = x^6$ **and**

$f(x) = x^7, f'(0) = f''(0) = f'''(0)$

$= f^{iv}(0) = f^{v}(0) = 0.$

$x = 0$ **is point of minima for** $f(x) = x^6$.

But $x = 0$ **is not point of maxima/minima for** $f(x) = x^7$

Hence, it is difficult to say anything.

Q.20 Answer:

has a local minimum

Solution:

$$f(0) = \sin 0 = 0, f\left(0^+\right) \to 0^+$$

$$f\left(0^-\right) = \lim_{x \to 0^-} \sin\left(x^2 - 3x\right) = \lim_{h \to 0} \sin\left(h^2 + 3h\right) \to 0^+$$

$$\text{Thus, } f\left(0^+\right) > f(0) \text{ and } f\left(0^-\right) > f(0)$$

Hence, $x = 0$ **is a point of minima.**

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